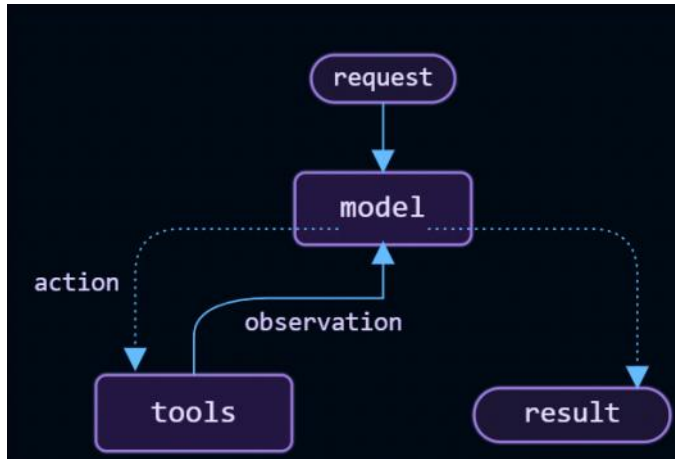


Agents - LangChain

Wednesday, July 29, 2026 7:55 PM

Agents: An agent is a model calling tools in a loop until a given task is complete.

An AI agent is smart system that can **understand a goal, make decisions, perform actions, observe the results**, and continue **until the goal is completed** with no human intervention.



Think of it this way:

- A **traditional program** follows fixed instructions.
- An **LLM (like ChatGPT)** generates text based on your prompt.
- An **AI agent** uses an LLM as its "brain" but can also **plan, use tools, remember information, and take actions**.

A Simple Real-Life Analogy: Imagine you want to book a vacation.

Normal Program:

You manually,

1. Search flights
2. Search hotels
3. Compare prices
4. Book everything

The software only does exactly what you click.

LLM: You ask: Suggest a 5-day trip to Goa.

It gives you an itinerary but can't actually book anything by itself.

AI Agent: You tell it: Plan a 5-day Goa trip under \$40,000

The agent:

1. Searches flights
2. Compares prices
3. Finds hotels
4. Check weather
5. Creates an itinerary

6. Asks for confirmation
7. Books everything.

It works toward the goal instead of just answering a question.

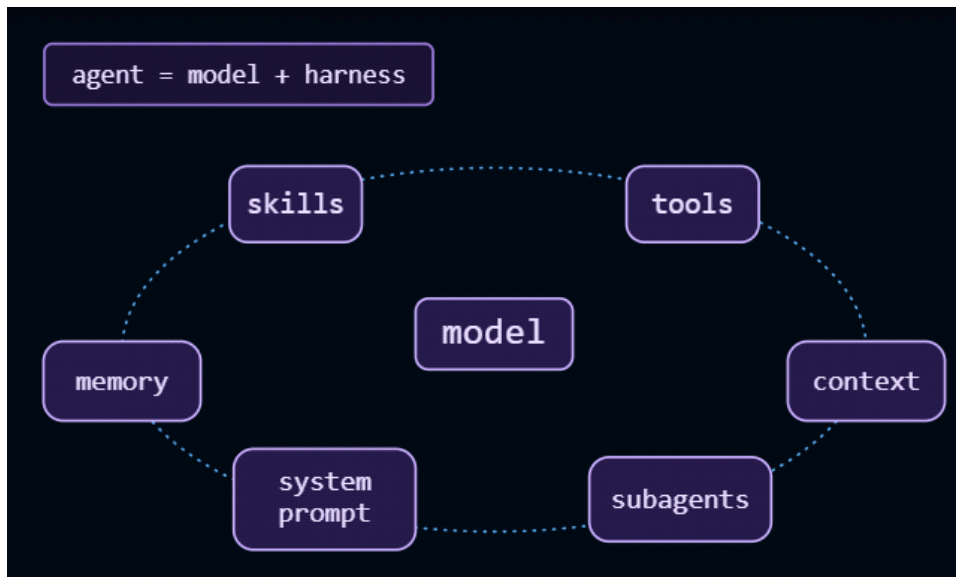
Different Between an LLM and an Agent:

LLM	AI Agent
Answers Questions	Completes goals
Produces text	Takes actions
One prompt -> One response	Multi step workflow
Doesn't plan much	Plans tasks
No built-in memory	Often maintains memory
Usually no external tools	Uses tools, APIs, DB, Browsers etc
Passive	Active

Core components:

This diagram explains the core building blocks of an AI agent.

Agent = Model + Harness



What does "Agent = Model + Harness" Mean?

Think of it like a human:

Model = Brain (LLM): the reasoning engine that understands language and decides what to do next.

Ex: GPT, Claude, Gemini, Mistral, Llama etc.

Analogy: The doctor's medical knowledge. Diseases, Treatments, Diagnosis
But can't diagnose a patient without more information.

Harness = Everything around the brain that helps it perform tasks.

Analogy: Patient History, Blood reports, X-rays, Medical instruments, Hospital staff, Previous records, Specialists etc.

Without these, even an expert doctor can't do the job effectively.

Doctor = Medical Knowledge + Hospital Resources

A brain alone can't send emails, search the internet, or remember previous conversations. Similarly, an LLM alone only predicts the next word.

AI agent = LLM + Tools + Memory, Context + Skills

Ex: memory, tools, instructions, content, specialized skills, other agents to collaborate with together, they form an AI agent.

Memory: Stores information from previous steps or interactions.

Ex: "The user prefers vegetarian food."

Planning: Breaks a complex goal into smaller tasks.

Ex: Goal: Create a company report.

Plan: Read docs, Extract facts, Summarize, Create slides, Email manager.

Tools: Agents become much more capable when they can use tools, such as:

- Search the web
- Read files
- Execute Python code
- Query databases
- Send emails
- Call APIs
- Interact with applications

Environment: The external world the agent interacts with, including files, websites, databases, devices or cloud services.

Example: Tell me today's weather.

LLM: I don't know today's weather unless I have current Information.

AI Agent:

- Uses a weather API.
- Retrieves today's forecast.
- Explains whether you need an umbrella.

Skills: Skills are reusable capabilities the agent has learned or been programmed with.

Examples: Think of skills as a person's professional expertise.

Customer support agent: answer FAQs, summarize complaints, classify tickets

Coding agent: explain code, debug programs, generate tests

Research agent: summarize papers, compare articles, create reports

Tools: Tools allow the agent to interact with the outside world.

Context: Context is the information the model currently uses to make decisions.

It includes: your prompt, uploaded documents, previous conversation, retrieved knowledge (RAG), instructions, task details.

Subagents: Sometimes one agent delegates work to other specialized agents.

Ex: Imagine building a travel plan.

Main Agent: "I need help!"

Flight Agent: Finds flights

Hotel Agent: Books hotels

Weather Agent: Check forecast

Budget Agent: Calculates costs

Main Agent : Combines everything into one plan

This is a multi-agent system, where each subagent focuses on a specific responsibility.

System Prompt: The system prompt defines the agent's role and behavior.

Ex:

You are a helpful medical assistant.

Never guess.

Always explain in simple language.

Cite reliable sources when appropriate.

This acts like the agent's job description and operating rules.

Memory: Memory lets the agent retain information beyond the current prompt.

There are two common types: Short-term-memory, Long-term-memory

Why are AI Agents Popular?

They can automate complex workflow such as:

- Customer support
- Research assistance
- Coding and debugging
- Financial analysis
- Medical document summarization with human oversight
- Personal assistants
- RAG systems
- Business process automation

Agent Frameworks: LangChain, LlamaIndex, Haystack, Microsoft AutoGen, CrewAI